

EXTREMAL GRAPH THEORY

Erdős Problem 915

Maximising Edge Density Subject to Connectivity Constraints

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The title, in plain words

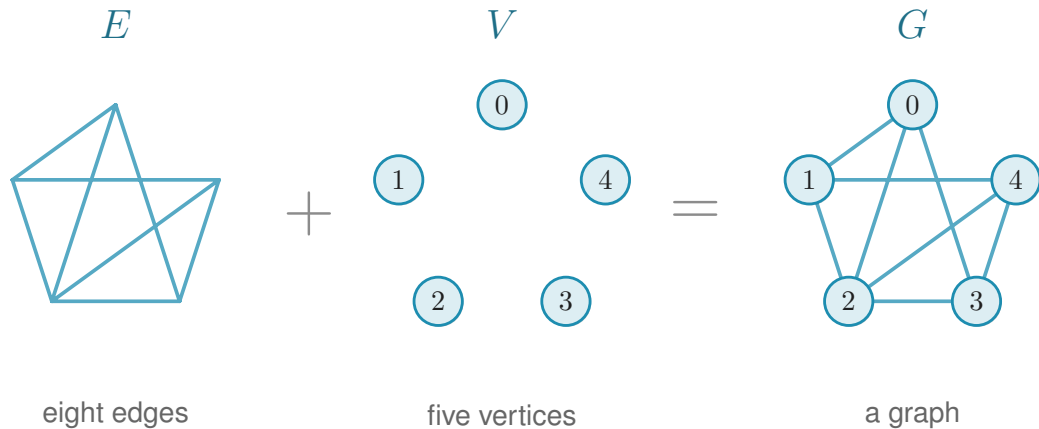
**Maximising edge density in graphs
under connectivity constraints**



translation

networks with many links and low connectivity

A graph is edges and vertices



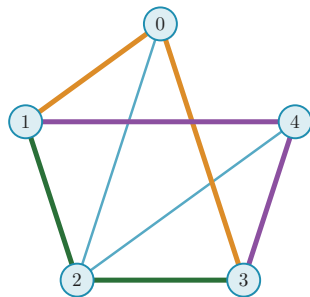
Erdős Problem 915

as many edges as possible, subject to the cap

$$\ell_m(n) = \max\{|E(G)| : \lambda_G^{\max} \leq m-1\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v) \text{ over every pair of vertices}$$

$\lambda_G(u, v)$ = the largest number of $u-v$ routes
that share **no edge**: one colour each



$$\lambda(1, 3) = 3$$

three routes from 1 to 3,
three colours, no edge reused

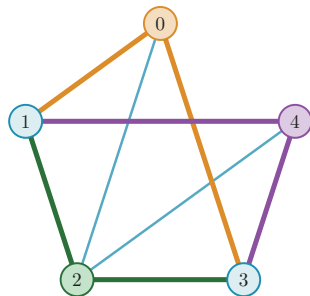
The same question, vertex by vertex

the same cap, read on vertices instead of edges

$$k_m(n) = \max \{ |E(G)| : \kappa_G^{\max} \leq m - 1 \}$$

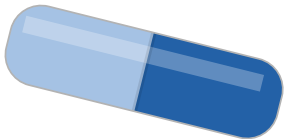
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v) \text{ over every pair of vertices}$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

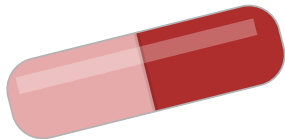
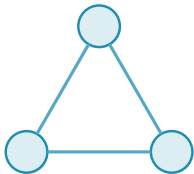


$$\kappa(1, 3) = 3$$

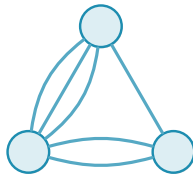
each route keeps its own interior
vertex: 0, 2 and 4

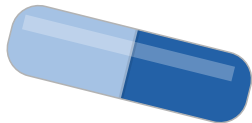


simple

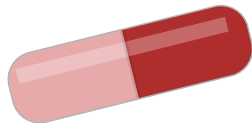
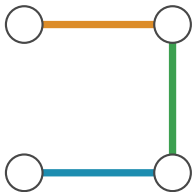


multi

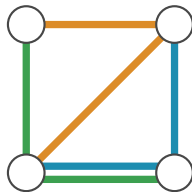


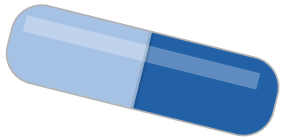


non-hypergraph

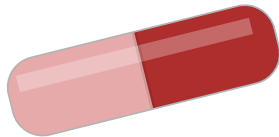
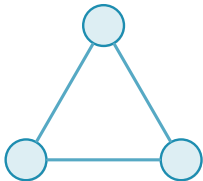


hypergraph

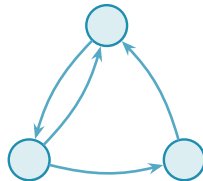




undirected



directed

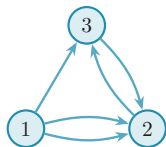


PART II

The program

Sixteen variants, one flow computation.

Modelling: two encodings



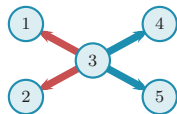
directed multigraph G

$\xrightarrow{\mu}$

$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$

multiplicity matrix μ

$\text{Diag}(\mu) = 0$ (no loops)



directed hypergraph H

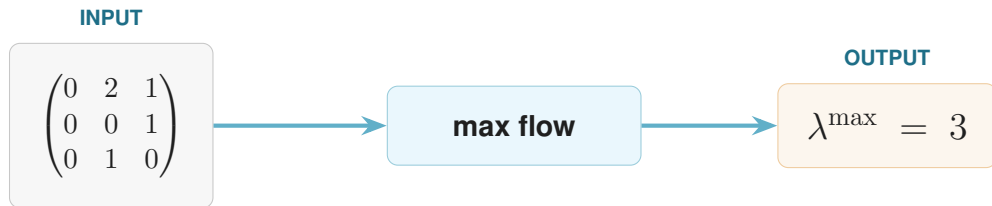
$\longrightarrow$

list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

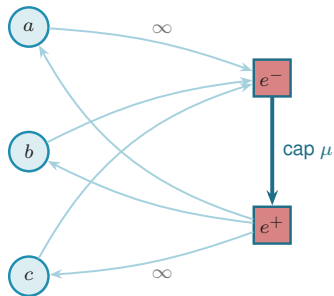
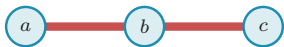
Measuring: one max-flow computation



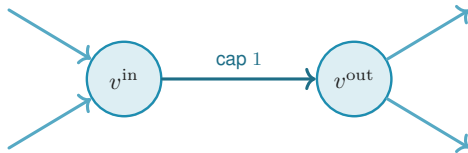
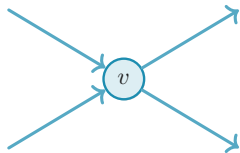
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph

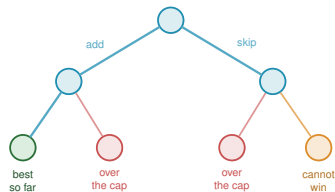


$$\kappa \rightarrow \lambda$$

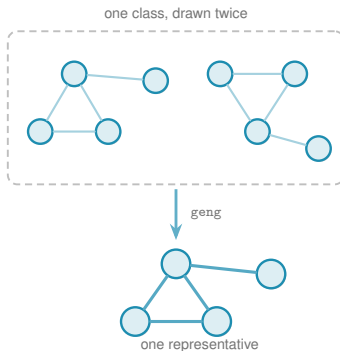


Efficiency: three ways to make the sweep finish

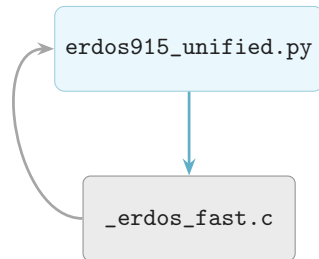
pruning



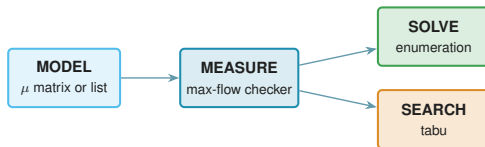
generating search



C: faster max flow



Solving versus searching



solving

- walk every graph on n vertices
- finishes $\Rightarrow$ the exact value
- times out $\Rightarrow$ a lower bound

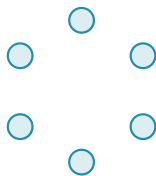
searching

VS

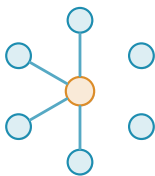
- steer by the energy Φ
- reaches sizes the sweep cannot
- always a lower bound

The search: one number per graph

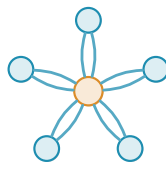
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \Lambda \text{ large, so breaking the cap never pays}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



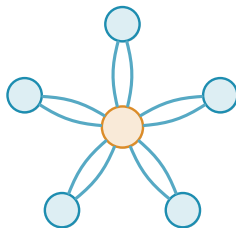
$$\Phi = -10$$

What the search finds on its own



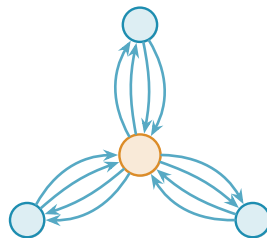
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

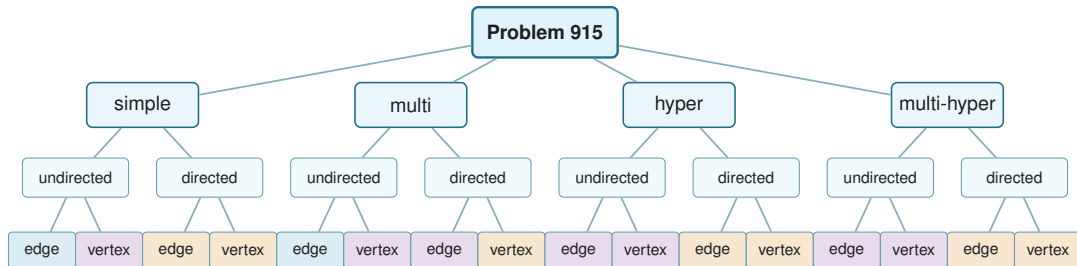
$$n = 6, m = 3, L_3(n) = 10$$



double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

Status of the sixteen variants



exact value established in the graph domain



exact value open, asymptotic bounds only



partial results or a conjectural general formula

AI: curse or blessing?

Pros

- $\text{\LaTeX}$ typesetting
- length & complexity of generated text

Cons

- length & complexity of generated text
- control

It's a tool.

short leash + AI proofs = argued conjecture [AI]

Thank you

`github.com/louisivandenbruwaene/thesis`

Simple graph · edge

$\ell_m(n)$

ESTABLISHED

$$\ell_m(n) = \begin{cases} \binom{n}{2}, & n < m, \\ \left\lfloor \frac{m(n-1)}{2} \right\rfloor, & n \geq m. \end{cases}$$

- Mader's upper bound, attained by hub-and-spokes for every $n \geq m$
- Gomory–Hu double count: at most $n - 1$ edges sit at tree distance one, every other edge is charged to two cuts

COMPUTED

Tabu search returns the tree at $n = 6, m = 2$: $5 = \ell_2(6)$.

$k_m(n)$

PARTIAL

$$k_m(n) = \ell_m(n) \quad (m \leq 4), \quad k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4 \quad (n \geq 14).$$

- at $m = 5$ the two agree through $n = 13$, and the first gap is 33 against 32 at $n = 14$
- $m \geq 6$ open since 1974. The block formulation would make it a knapsack over feasible 2-connected blocks with rate $c_m = \lim_n k_m(n)/(n-1)$, and $m/2 < c_m \leq 2(m-1)$ under it

COMPUTED

Finite block sweeps inspect small blocks only, and the recursive constructions in the literature show why they cannot fix c_m .

Simple digraph · arc

$\ell_m^{\text{dir}}(n)$

OPEN

$$\ell_2^{\text{dir}}(n) = M(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right), \quad \ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n).$$

- upper $n^2/4 + (m-1)n + O_m(1)$ for large n , from Huang and Lyu at $t = m-1$, and this thesis adds an elementary bound valid at every n
- lower $n^2/4 + (m-2)n/2 + O_m(1)$, the augmented bipartite digraph
- the exact value and the linear coefficient are open, and the clique-core example disproves the earlier hub-versus-bipartite formula

COMPUTED

Enumeration confirms $m = 2$ through $n = 7$. Tabu returns 6, 10 and 12 at $n = 4, 6$ and 7 .

Simple digraph · vertex

$$k_m^{\text{dir}}(n)$$

OPEN

$$k_2^{\text{dir}}(n) = M(n), \quad k_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n) \quad (m \geq 3).$$

- the $m = 2$ argument never mentions the separation, so it runs for κ and the arc value follows from it
- whether the two agree for $m \geq 3$ is open: an arc upper bound does not constrain the larger vertex-feasible family

COMPUTED

Enumeration confirms $m = 2$ through $n = 7$, and the two separations agree in the completed $m = 3$ runs through $n = 6$.

Multigraph · edge

$L_m(n)$

ESTABLISHED

$$L_m(n) = (m - 1)(n - 1).$$

- a tree with every edge repeated $m - 1$ times attains it
- the upper bound counts cuts, so it holds whatever shape a feasible multigraph has

COMPUTED

Tabu search returns the multiplicity-2 star at $n = 6$, $m = 3$: $10 = L_3(6)$.

$K_m(n)$

PARTIAL

$$K_2(n) = n - 1, \quad K_3(n) = 2(n - 1) \text{ (proposed).}$$

- parallel copies are separated routes under the incidence convention, so this is not the simple problem under another name: $K_5(4) = 14$ against $L_5(4) = 12$
- the proposed block reduction gives order m^2n , with coefficients between $3 - 2\sqrt{2}$ and 2 and neither a common limit nor the best blocks established

COMPUTED

Exhaustion gives $K_5(4) = 14$ and $K_5(5) = 19$. A nine-vertex sweep over all 191826 two-connected graphs gives $g_6(9) = 54$.

Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

PARTIAL

$$L_m^{\text{dir}}(n) = (m - 1)M(n), \quad M(n) = \max(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor).$$

- one reachability-preserving subgraph is removed per unit of $m - 1$, rather than a vertex at a time
- the fractional optimum is $M(n)$ too, so one m -free optimisation bounds every threshold at once
- the classification of the extremisers for $n \geq \max(8, 2m)$ stays conjectural

COMPUTED

Tabu returns the double star at $m = 3$: 12 at $n = 4$ and 16 at $n = 5$.

Directed multigraph · vertex

$K_m^{\text{dir}}(n)$

OPEN

$$K_m^{\text{dir}}(n) = (m - 1)\frac{n^2}{4} + O_m(n) \quad (\text{proposed}).$$

- the arc extremiser is not even a finite optimum here: $K_6^{\text{dir}}(3) = 24$ against the established arc maximum of 20
- the exact correction and the most effective block shapes are open

COMPUTED

Direct enumeration gives $K_6^{\text{dir}}(3) = 24$.

Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

PARTIAL

$$\ell_m^{(r)}(n) \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor.$$

- cut induction for the bound, and attainment established on $m-1 \leq \binom{n-2}{r-2}$
- off that range the supply of distinct hyperedges runs out: at $n = r = m = 3$ the bound reads 2 and only one hyperedge exists

COMPUTED

Exhaustive enumeration gives 11 at $(n, m, r) = (6, 6, 3)$, below the bound 12.

Simple hypergraph · vertex

$$k_m^{(r)}(n)$$

PARTIAL

$$k_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor, \quad k_3^{(r)}(n) \leq \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor.$$

- $m = 2$ is feasible exactly when the incidence graph is a forest, and a star hypertree attains the count
- $m = 3$ comes from the incidence-rank inequality, with simple attainment for $2 < r < n$
- $m \geq 4$ open: $\kappa \leq \lambda$ carries an edge bound the wrong way, since vertex feasibility is the weaker restriction

COMPUTED

No counterexample at $m = 4$ across eleven (n, r) cells. The $r = 2$ case of the same problem fails at $m = 5$.

Multihypergraph · edge

$$L_m^{(r)}(n)$$

PARTIAL

$$L_m^{(r)}(n) \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor.$$

- the cut bound is unchanged by repetition, the constructions are not
- a star hypertree with every edge repeated $m-1$ times reaches $(m-1)(n-1)/(r-1)$ whenever $(r-1)$ divides $n-1$
- at $n=r=m=3$, two copies of the only triple reach a value the simple model cannot

COMPUTED

A twelve-copy construction at $(n, m, r) = (6, 6, 3)$ exceeds the simple edge optimum of 11.

$$K_m^{(r)}(n)$$

PARTIAL

$$K_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor, \quad K_3^{(r)}(n) \leq \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor.$$

- repetition does not help at $m = 2$: two copies of one hyperedge are already two separated routes
- at $m = 3$ the bound is the one above, and the repeated star supplies further attainment cases
- larger thresholds open

COMPUTED

The twelve-copy edge witness at $(6, 6, 3)$ is a vertex lower bound, and does not prove the vertex maximum is twelve.

Directed simple hypergraph · arc

$$\ell_m^{\text{dir}(r)}(n)$$

OPEN

$$\ell_m^{\text{dir}(r)}(n) = (1 + o(1)) \frac{(m-1)n^2}{4(r-1)} \quad (r \geq 3).$$

- a shadow digraph on tail–head pairs, with a matching argument for the distinct two-step routes that share one hyperedge
- attained asymptotically by a two-layer construction whose heads have bounded incidence degree
- finite exact values open

COMPUTED

At $n = m = 6$, $r = 3$, giving each tail five distinct outgoing hyperedges produces 30 of them, with a direct degree-cut feasibility check.

Directed simple hypergraph · vertex

$$k_m^{\text{dir}(r)}(n)$$

OPEN

$$k_m^{\text{dir}(r)}(n) = (1 + o(1)) \frac{(m-1)n^2}{4(r-1)} \quad (r \geq 3).$$

- the shadow argument re-runs under the stricter separation, so the leading term is the same
- whether the finite edge and vertex maxima agree is open

COMPUTED

Random greedy growth with shuffled restarts measures what the implemented search finds, and establishes no optimum.

$$L_m^{\text{dir}(r)}(n)$$

OPEN

$$L_m^{\text{dir}(r)}(n) = (1 + o(1)) \frac{(m-1)n^2}{4(r-1)} \quad (r \geq 3).$$

- repetition removes the distinct-head-set restriction in the two-layer construction, so the finite values can differ from the simple model
- forward, backward and general orientations share this leading term
- at $r = 2$ these are directed multigraphs, and the simple $r = 2$ coefficient is $1/4$ instead, which is why $r \geq 3$ is kept

COMPUTED

The plotted experiments use rank three and the forward convention. General orientations are checked separately in the appendix.

$$K_m^{\text{dir}(r)}(n)$$

OPEN

$$K_m^{\text{dir}(r)}(n) = (1 + o(1)) \frac{(m-1)n^2}{4(r-1)} \quad (r \geq 3).$$

- the same established leading term under both separations
- finite exact values open, and whether forward and general agree once $n > r$ is open

COMPUTED

The rank-three forward-convention runs are not an exhaustive comparison of the three orientation models.